Opioid-Related ED Cohort Pipeline (opioid_ed)

Phase 0: Virginia APCD Longitudinal Lake (2016–2020)
Medical Claims Lake
(All-Payer Encounters, Diagnosis & Procedure Positions 1–10)
Pharmacy Claims Lake
(Adjudicated Dispensing, NDC Mappings, Days Supply)
Annual Partitions (2016, 2017, 2018, 2019)
Stratified into 8 Age Bands (0–12 to 85–114)
Claims Stream
Enrollment Roster
Phase 1: Case Identification & Exclusivity (phase1_data_preparation.py)
10-Position Diagnostic Scanner
Scans Primary through 10th ICD Position
Qualifying Emergency Encounter
Opioid ED Target Cases (t_0)
• ICD-10 F11.20 (Opioid Dependence)
• Acute Poisonings: T40.0X–T40.4X
Timestamp: first_opioid_ed_date
Validate Exclusivity
Cohort Exclusivity Boundary
Strictly drops patients meeting
Polypharmacy ED criteria
Phase 3: 1:5 Matched Case–Control Sampling (phase3_cohort_creation.py)
Eligible Control Pool
• Covered APCD members in same partition year
• Zero opioid-related diagnoses in study year
• Verified health system contact continuity
Candidate Pool
1 Target Case
1:5 Incidence-Density Matching (No Replacement)
Exact Stratification On:
1. Age Band (8 Strata: 0–12, 13–24, 25–44, ..., 85–114)
2. Biological Sex (Exact M/F Match)
3. Calendar Year Partition (2016–2019)
4. Active Coverage Continuity
Assign Anchor
Pseudo-Index Reference Date (t_pseudo)
Controls assigned earliest non-ED medical claim date
Guarantees identical 12-month baseline window
Validated Cases
5 Matched Controls
Phase 2: Temporal Exposure & Leakage Audit (phase2_event_processing.py)
12-Month Pre-Index Baseline Window
[t_0 - 365 days → t_0 - 1 day]
• Longitudinal Opioid Titration & MME Escalation
• Sedative Overlap (Benzodiazepines, Gabapentinoids)
• CPIC CYP2D6/CYP2C19 Substrate Burden
Audit All Features
Step 3b Temporal Reverse-Causality Audit
Purges 348 Post-Index Treatment Codes:
• 164 OUD Meds (Buprenorphine, Naloxone, Vivitrol)
• 84 Diagnostic CPTs (CPT 80307 Urine Drug Screens)
• 100 Hospital Post-Stabilization Billing Codes
Lock Clean Features
Locked 682-Feature Whitelist
Strictly derived from pre-index claims
Applied symmetrically across Cases & Controls
Tabular Feature Matrix
Phase 4: QA Assertion & Checkpointed Export (phase4_finalization.py)
DuckDB QA Unit Assertion (qa_index_date_uniqueness.py)
SELECT mi_person_key, count(DISTINCT first_opioid_ed_date)
HAVING count > 1;
Asserts EXACTLY ONE index date per person-year
QA Passed
Cell-Size Suppression Guard
Suppresses n ≤ 10 cell counts per CMS/VHI privacy policy
Write Gold Artifact
Final Parquet Fact Table (s3://pgx-repository/cohorts/)
• schema: final_cohort_schema.json
• Cases + 5:1 Controls with 682 Whitelisted Features
• Checkpointed to S3 pgx-pipeline-status/